

Individual Progress Report

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Group Topic: **Urban Air Quality & Environmental Health**

Lab No. Week 3

Git Repo/ Colab Link: <https://github.com/Cristy16/BDA---Final-Project> Date: 05/08/26

Objectives

General Objective: To select, implement, fine-tune, and evaluate a machine learning algorithm on the validated Urban Air Quality dataset. The group chose Prediction (Regression) as the analytical framework, using a Random Forest Regressor to estimate PM2.5 concentrations from CO, NO2, SO2, and location features.

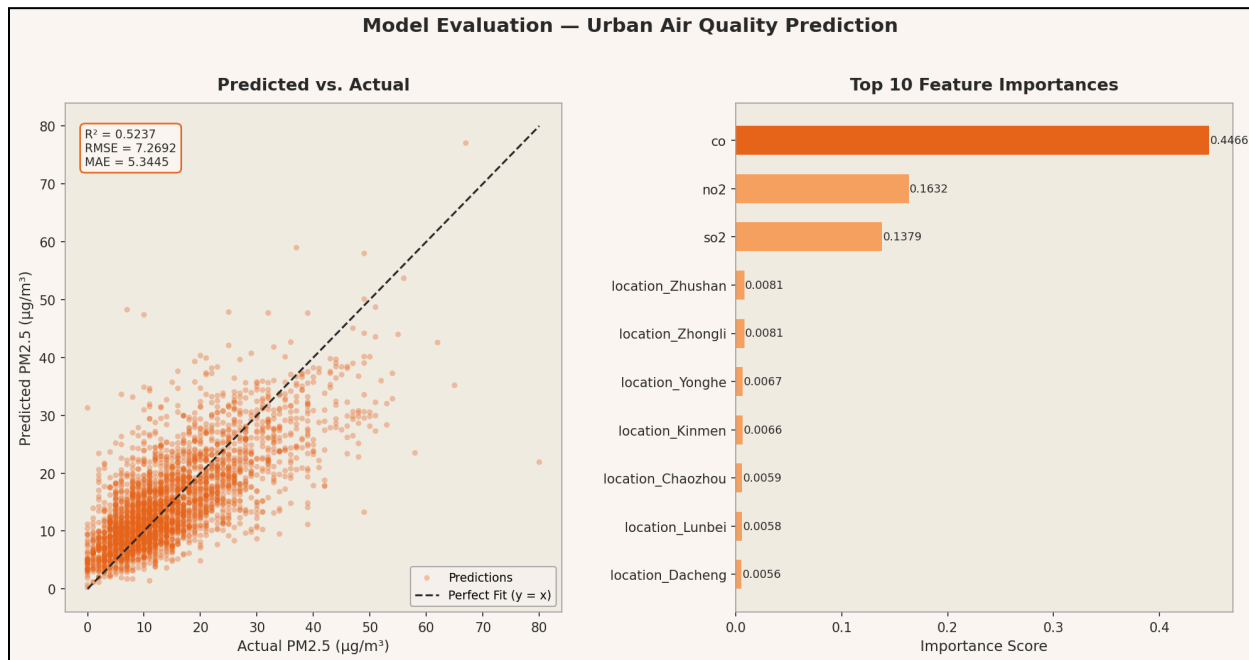
Individual Objective (Evaluation and Visualization): To evaluate the final tuned model's predictive performance using standard regression metrics and to produce visualizations that communicate the results clearly. Specifically:

- Compute R^2 , RMSE, and MAE on the withheld 134,972-row test set using the tuned Random Forest Regressor.
- Produce a Predicted vs. Actual scatter plot and a Feature Importance bar chart.
- Make the Evaluation & Visualization section of the team Jupyter Notebook, including data ethics documentation.

Tasks Completed

1. Reviewed the full Week 3 notebook produced by Ian, Danilyn, Polly, and Erick — confirming the complete pipeline from data loading and preprocessing, through baseline training and fine-tuning, before authoring the evaluation phase.
2. Authored the Evaluation & Visualization section of the team Jupyter Notebook — added a Markdown documentation cell (metric definitions table, visualization descriptions, and data ethics notes) and a Python code cell implementing all evaluation and charting logic.
3. Computed final regression metrics on the 134,972-row test set using `y_test` and `y_pred_tuned` from Erick's fine-tuning output. The tuned model (best params: `n_estimators=50`, `min_samples_split=10`, `max_depth=None`, `bootstrap=True`) outperformed the baseline across all metrics:
 - $R^2 = 0.5237$ (vs. baseline 0.4893)
 - RMSE = 7.2692 $\mu\text{g}/\text{m}^3$
 - MAE = 5.3445 $\mu\text{g}/\text{m}^3$
4. Produced two visualizations saved as `phase7_evaluation.png`:

- Predicted vs. Actual scatter plot with perfect-fit reference line and embedded metrics annotation box.
- Top 10 Feature Importances horizontal bar chart ranked by Gini importance score from the tuned Random Forest.



Tasks to be Completed

I was able to complete all task assignments given to me this Week 3 activity, which is the Evaluation & Visualization phase. The next step is to finalize and push the complete notebook to the team Git repository, which will be handled by the member assigned to that task.

Results and Analysis

The tuned Random Forest Regressor was evaluated on the withheld 134,972-row test set. Final metrics:

Metric	Value
R^2 (R-squared)	0.5237
RMSE	7.2692 $\mu\text{g}/\text{m}^3$
MAE	5.3445 $\mu\text{g}/\text{m}^3$
Test Set Size	134,972 rows

Training Set Size	539,884 rows
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- The model achieved an R^2 of 0.5237, meaning it explains approximately 52% of the variance in PM2.5 concentrations using CO, NO2, SO2, and location as predictors. This is a clear improvement over the baseline model's R^2 of 0.4893, confirming that Erick's RandomizedSearchCV fine-tuning step was effective.
- The RMSE of 7.2692 $\mu\text{g}/\text{m}^3$ indicates the average prediction error is roughly 7.27 $\mu\text{g}/\text{m}^3$ — a meaningful level of precision given PM2.5 values range from near 0 to 1,000 $\mu\text{g}/\text{m}^3$. The MAE of 5.3445 $\mu\text{g}/\text{m}^3$ further confirms stable performance across the bulk of the test set.
- The Predicted vs. Actual scatter plot shows predictions clustering tightly around the perfect-fit line in the low-to-moderate PM2.5 range, with increasing spread at extreme pollution spike values — consistent with the anomalous events identified in Phase 3 and a known limitation of the current feature set.
- The Feature Importance chart confirms CO as the dominant predictor, followed by NO2 and SO2, consistent with the 0.78 CO–NO2 correlation identified in Phase 2. Location-encoded features contribute secondary importance. The remaining ~48% of unexplained variance is attributed to factors outside the dataset (weather, seasonality, traffic volume), documented as a known limitation.

Challenges and Solutions

Challenge: A NameError for `y_test` occurred when attempting to run the evaluation cell in isolation — the variable did not exist in memory because the upstream cells had not yet been executed in the current session.

Solution: Used “Run All Cells” from the VS Code top menu to execute the full notebook from top to bottom, ensuring all upstream variables (`y_test`, `y_pred_tuned`, `tuned_model`, `X_test`) were loaded into memory before the evaluation cell ran.

Conclusion

All Week 3 objectives were met. The team successfully selected, implemented, fine-tuned, and evaluated a Random Forest Regression model for PM2.5 prediction. As Evaluation & Visualization Lead, the final phase of the notebook is complete with computed metrics ($R^2=0.5237$, $\text{RMSE}=7.2692$, $\text{MAE}=5.3445$), two production-quality visualizations, and a data ethics review. The tuned model outperforms the baseline and its limitations are transparently documented.